

Python Basics Assignment

Topics: print(), Variables, Data Types, Arithmetic Operators, Logical Operators

1. Student Information

Create variables for a student's name, age, city, and student status. Print all the information in a properly formatted output.

2. Data Types

Create variables containing a string, integer, float, and boolean. Print the value and data type of each variable using type().

3. Arithmetic Operations

Create two number variables and calculate their:

- Addition
- Subtraction
- Multiplication
- Division
- Modulus
- Power

4. Student Marks Calculator

Store the marks of three subjects in separate variables. Calculate and print the **total marks** and **average marks**.

5. Shopping Bill

Create variables for the product price and quantity. Calculate and print the total bill.

6. Rectangle Area

Create variables for the length and width of a rectangle. Calculate and print its **area** and **perimeter**.

7. Time Converter

Store 250 minutes in a variable. Calculate and print the number of **hours** and **remaining minutes**.

8. Student Eligibility

Create variables for marks and attendance.

A student is eligible if:

- Marks are greater than or equal to 50
- Attendance is greater than or equal to 75

Use logical operators to calculate and print whether the student is eligible.

9. Logical Operators

Create two Boolean variables:

a = True

b = False

Print the results of:

- a and b
- a or b
- not a
- not b

Final Challenge Student Report

Create variables for:

- **Student name**
- **Age**
- **Marks of 3 subjects**
- **Attendance**

Then:

1. **Print the student's information.**
2. **Calculate and print the total marks.**
3. **Calculate and print the average marks.**
4. **Use logical operators to calculate and print whether the student is eligible:**
 - **Average marks >= 50**
 - **Attendance >= 75**

Restriction: Do not use if, else, or elif

